



2014

TUTORIAL

T6:
ANALOG FRONT-END DESIGN
FOR G_b/S WIRELINE RECEIVERS

INSTRUCTOR: ELAD ALON

February 9th, 2014

Acknowledgements

- These slides include material from:
 - Prof. Vladimir Stojanovic (UC Berkeley)
 - Jared Zerbe (Apple)
 - Prof. Borivoje Nikolic (UC Berkeley)

Selected References

- [1] J. Zerbe et al., "Equalization and Clock Recovery for a 2.5 – 10Gb/s 2-PAM/4-PAM Backplane Transceiver Cell," *IEEE JSSC*, Dec. 2003
- [2] V. Stojanovic et al., "Modeling, Analysis, and Design of High-Speed Links," Ph.D. thesis, Stanford University, Dec. 2004.
- [3] S. Shekhar et al., "Design Considerations for Low-Power Receiver Front-End in High-Speed Data Links," *IEEE CICC*, Sept. 2013
- [4] C. Thakkar et. Al, "A 10Gb/s 45mW Adaptive 60GHz Baseband in 65nm CMOS," *IEEE JSSC*, Apr. 2012
- [5] M. Park et. al., "A 7Gb/s 9.3mW 2-Tap Current-Integrating DFE Receiver," *IEEE ISSCC*, Feb. 2007
- [6] B. Leibowitz, et al., "A 7.5Gb/s 10-Tap DFE Receiver with First Tap Partial Response, Spectrally Gated Adaptation, and 2nd-Order Data-Filtered CDR," *IEEE ISSCC*, Feb. 2007
- [7] K. K. Parhi, "High-speed architectures for algorithms with quantizer loops," *IEEE ISCAS*, May 1990
- [8] H. Wang et al., "A 21-Gb/s 87-mW transceiver with FFE/DFE/linear equalizer in 65-nm CMOS technology," *IEEE JSSC*, Apr. 2010

T6: ANALOG FRONT-END DESIGN FOR Gb/s WIRELINE RECEIVERS



This tutorial introduces the design, analysis, and optimization of analog front-ends for state-of-the-art Gb/s wireline receivers. Specifically, the tutorial highlights key circuit/system requirements, tradeoffs, and gotchas inherent in the design of such receivers. In addition, the tutorial develops simple analytical formulas that accurately predict the power and performance characteristics of key circuit blocks such as continuous-time linear equalizers and decision-feedback equalizers.

Instructor: Elad Alon

Elad Alon received the Ph.D. degree in Electrical Engineering from Stanford University, Stanford, CA, in 2006. In 2007, he joined the University of California, Berkeley, where he is now an Associate Professor of Electrical Engineering and Computer Sciences, as well as a Co-Director of the Berkeley Wireless Research Center (BWRC). He has held consulting or visiting positions at Cadence, Xilinx, Wilocity, Oracle, Intel, AMD, Rambus, Hewlett Packard, and IBM Research. His research focuses on energy-efficient integrated systems, including the circuit, device, communication, and optimization techniques used to design them. He received the IBM Faculty Award in 2008, the 2009 Hellman Family Faculty Fund Award, and the 2010 UC Berkeley Electrical Engineering Outstanding Teaching Award, and has co-authored papers that received the 2010 ISSCC Jack Raper Award for Outstanding Technology Directions Paper, the 2011 Symposium on VLSI Circuits Best Student Paper Award, and the 2012 CICC Best Student Paper Award.

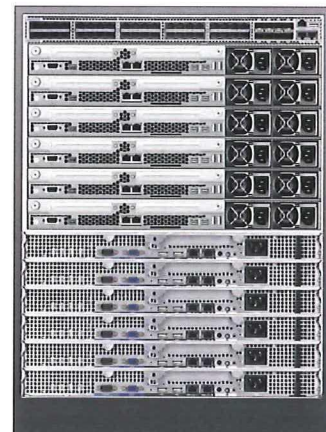
Analog Front-End Design for Gb/s Wireline Receivers

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Importance of Wireline Links



Images from eetimes.com

- Wireline I/O performance and power increasingly critical differentiators
 - Chassis can contain tens of 1000's of these links

Key Take-Home Lessons (2)

- Minimize dynamic range on the high-speed signal path
 - Watch out for block ordering
 - Keep DACs for e.g. offset cancellation, channel adaptation off the signal path
- Think about the whole system!
 - Often opportunities to save power by moving functionality/problems from one place to another

ISSCC 2014 Papers to Check Out

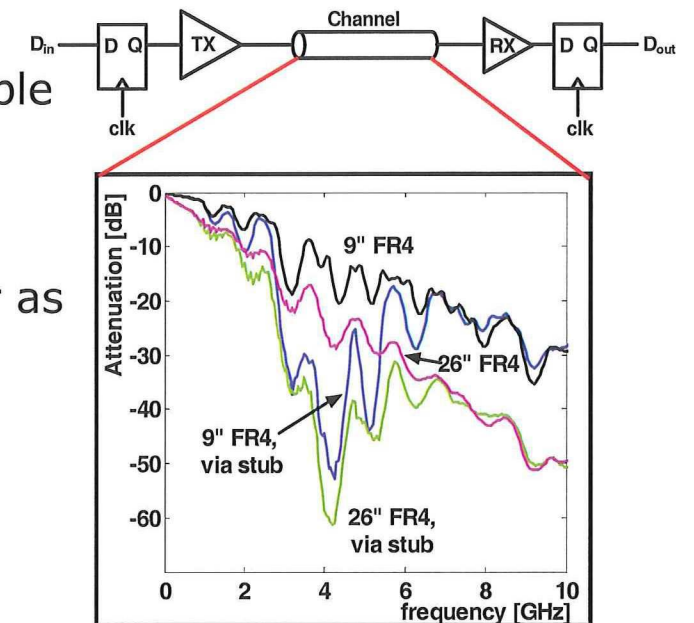
- Papers 2.1, 26.1: Front-end design issues within state-of-the-art 20 - 28Gb/s transceivers
- Papers 2.4, 2.5: Extremely energy-efficient 16-25Gb/s front-ends based on current-integration and charge domain techniques

Tutorial Outline

- 1 RX AFE Basics
- 2 RX AFE Circuits & Power Analysis
- 3 Practical Issues and Common Gotchas
- 4 Key Lessons

The High-Speed Wireline Challenge

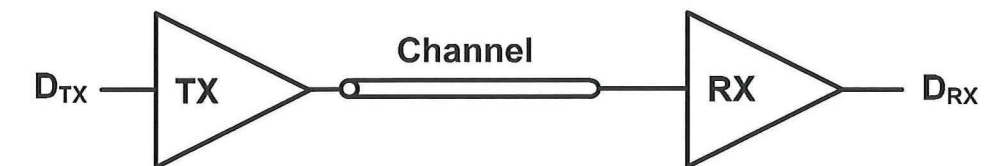
- Deliver as much throughput as possible
 - Multiple Tb/s for the whole chip
- Using as little power as possible
 - $\sim 1\text{pJ/bit}$ or less
- While dealing with "badness" of the channels



Key Take-Home Lessons (1)

- Receiver AFE circuit power easy to predict
 - Calculate power early and often – feed back to system design
- Analog circuits themselves can be very low power
 - Even for many taps (DFE)
 - Power often dominated by large number of high-speed digital structures (flops)

Focus of this Tutorial: RX

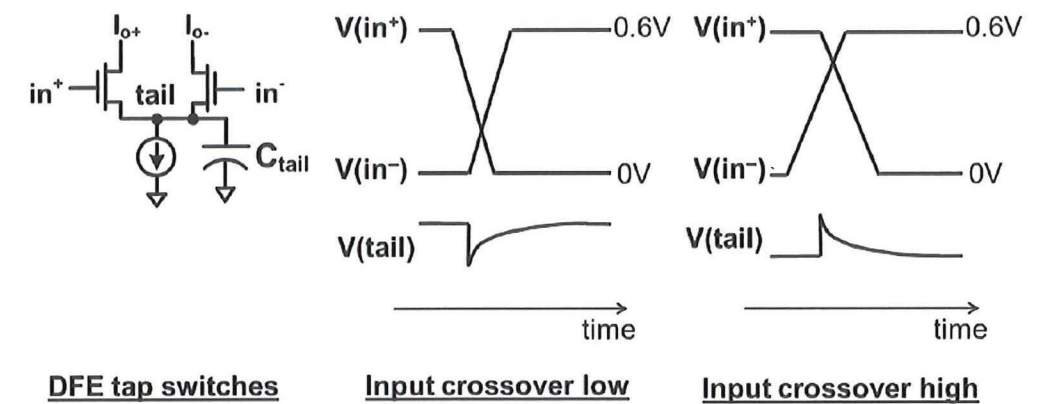


- By definition, TX knows the data, RX doesn't
 - Tends to place more stringent constraints on RX circuit design
- RX impedance levels tend to be higher – more efficient to implement signal processing there
- Knowledge of RX design techniques maps directly to TX design too

Tutorial Outline

- RX AFE Basics
- RX AFE Circuits & Power Analysis
- Practical Issues and Common Gotchas
- Key Lessons

Impact of Tap Tail Capacitance

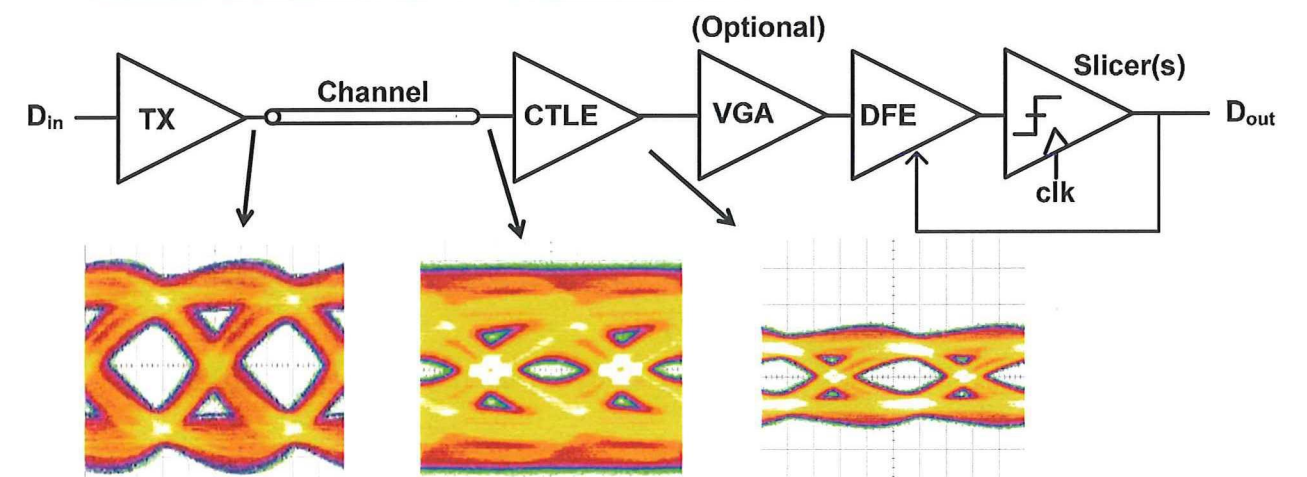


- Leads to signal-dependent glitches in DFE response
- Good news: adaptation can & will correct these errors^[4]

Tutorial Outline

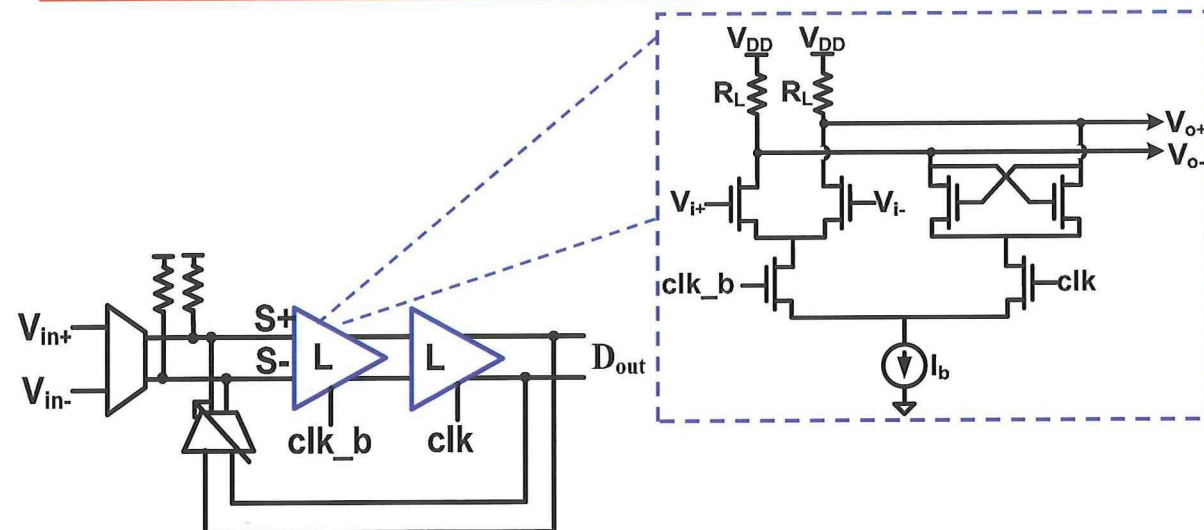
- RX AFE Basics
 - Basic AFE requirements
 - RX Equalizers
- RX AFE Circuits & Power Analysis
- Practical Issues and Common Gotchas
- Key Lessons

Complete Signal Path: Linearity



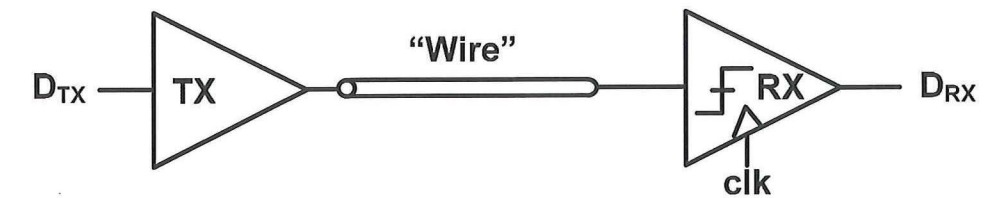
- Ordering of components on RX matters
 - Especially in systems with large TX swing
- CTLE should always be **first**
 - Reduces dynamic range caused by ISI

Option 3: All CML Design^[8]



- Reduces required comparator gain
- [9] demonstrated over 20Gb/s in 65nm CMOS

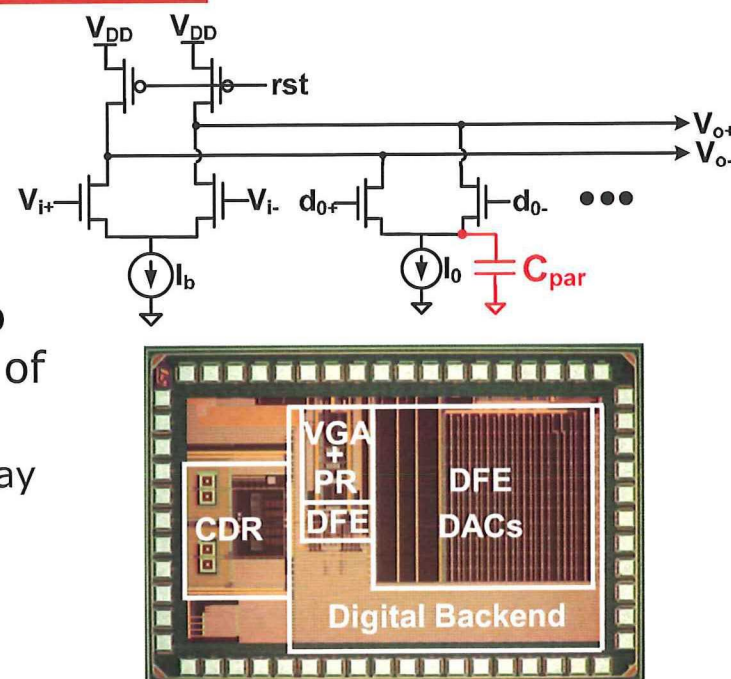
Most Basic RX



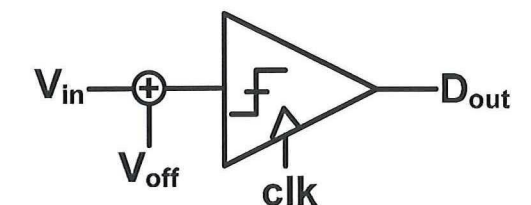
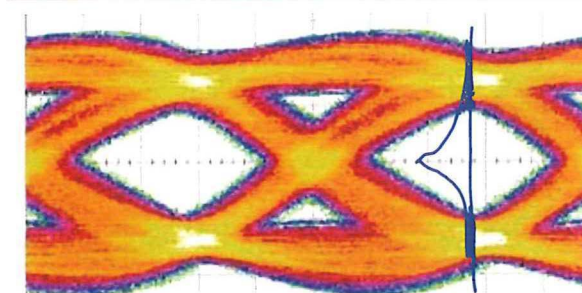
- Most basic RX really is "just" a comparator
 - Converts potentially small, analog voltages on the "wire" into digital levels
- What can go wrong?

DFE Physical Design

- Tap DACs must be high resolution, hence large
- Don't want DACs to increase size (cap) of signal path
 - DACs placed far away
- Result: large C_{par} ...



Key Constraint: Bit Error Rate

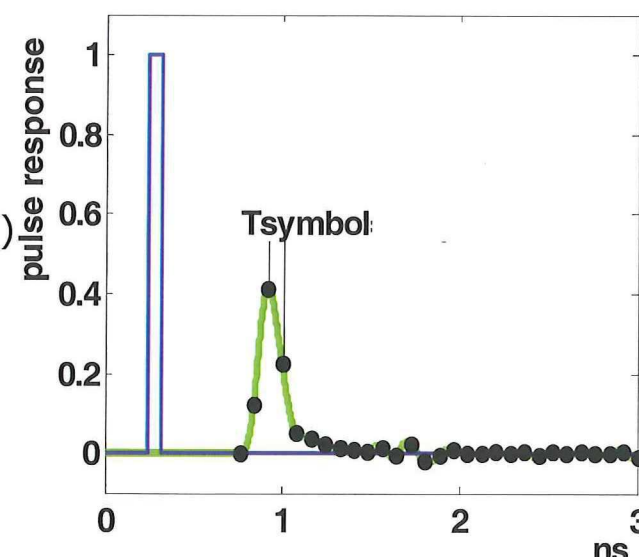


- BER = Bit Error Rate
 - Average # of wrong received bits / total transmitted bits
- Simplified example: (voltage only)

$$BER = \frac{1}{2} \operatorname{erfc} \left(\frac{V_{in, ampl} - V_{off}}{\sqrt{2} \sigma_{noise}} \right)$$
- $BER = 10^{-12}$: $(V_{in, ampl} - V_{off}) = 7\sigma_n$
- $BER = 10^{-20}$: $(V_{in, ampl} - V_{off}) = 9.25\sigma_n$

Noise Not the Only Source of Errors: ISI

- ISI: Inter-Symbol Interference
- Channel is band-limited
 - I.e., dispersive (low pass)
 - Short TX pulses get spread out
 - Low latency
- Also get reflections
 - Z mismatches, connectors, etc.
 - Longer latency

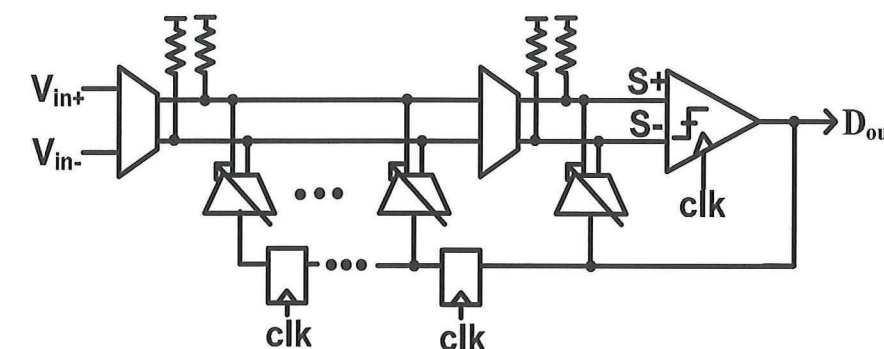


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9

Option 1: Split Summation^[6]



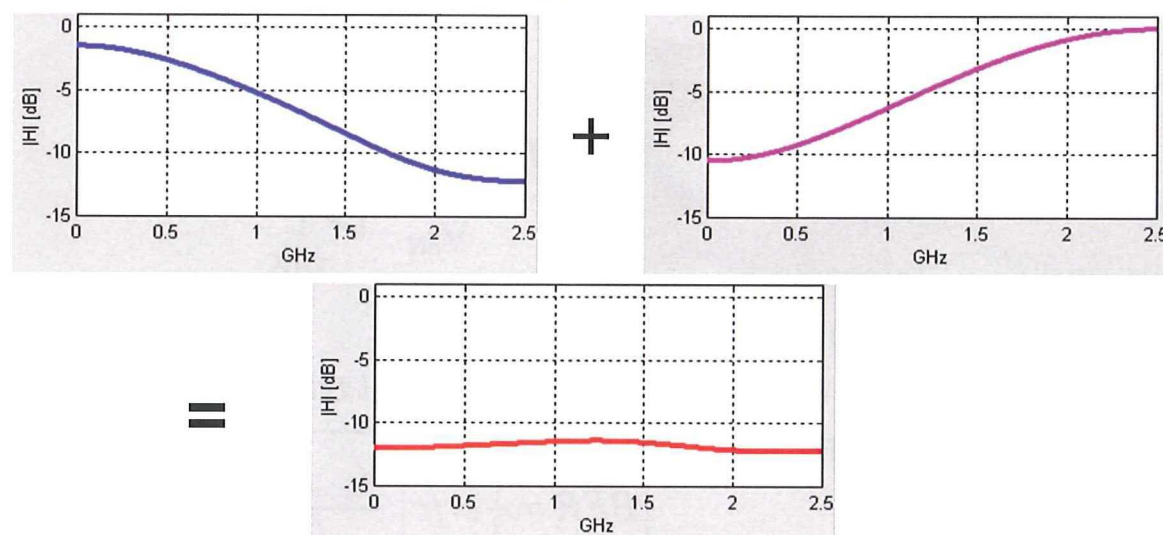
- Faster summer (less self-loading) for 1st tap
- Watch out for non-linearity
 - First summer (later taps) may distort the signal

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43

Equalization



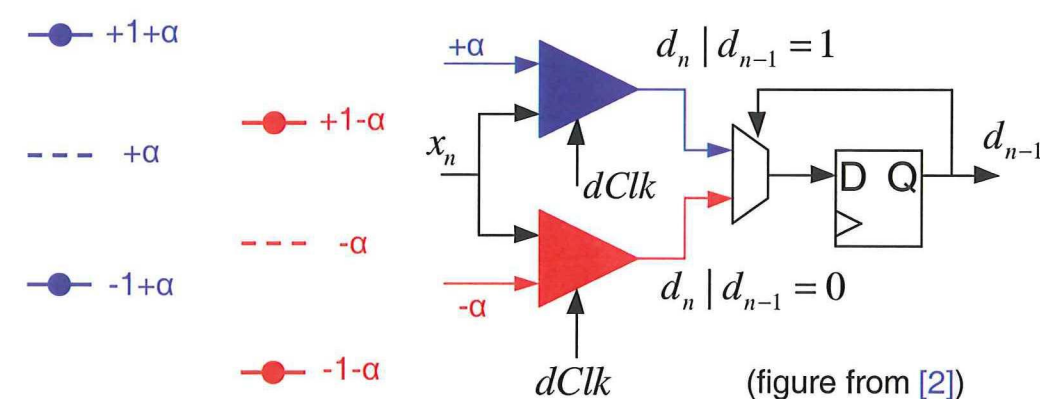
- Basic goal is to “flatten” channel response
 - I.e., in time domain, get back our nice clean pulse

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Option 2: Loop Unrolling^[7]



- Pushes feedback loop into digital domain
- Watch out for later taps (if there are any) though
 - MUX adds extra latency

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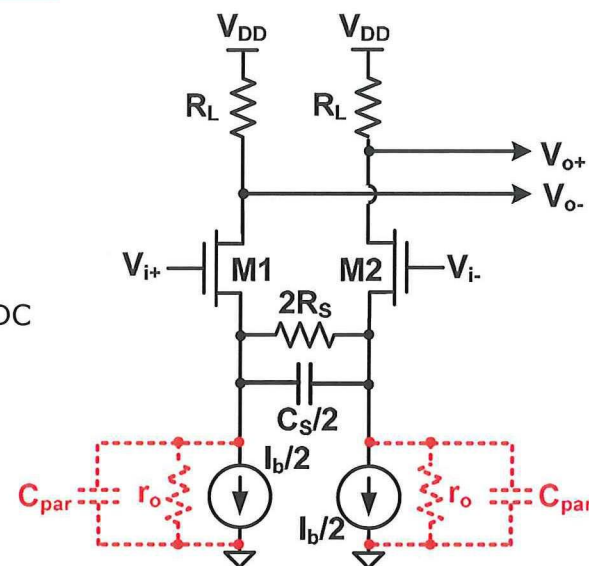
CTLE Design Issues (2)

Cause:

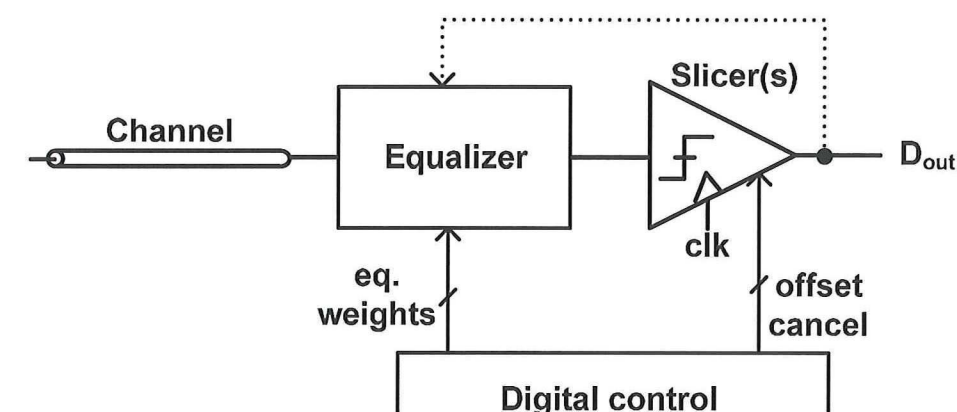
- Tail device r_o , C_{par}

Effects:

- r_o limits maximum A_{pk}/A_{DC}
- C_{par} limits maximum ω_p
 - I.e., pole @ source associated with ω_{zero}

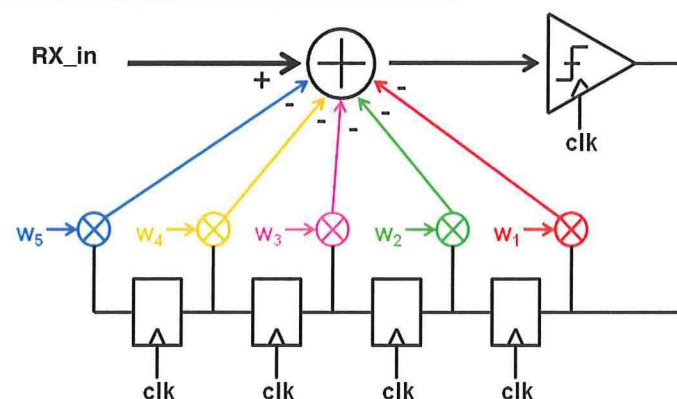


More Complete RX AFE



- Let's look at the types of equalizers we will need
 - Note that equalization is also often done on TX
 - But equalization burden is increasingly moving to RX

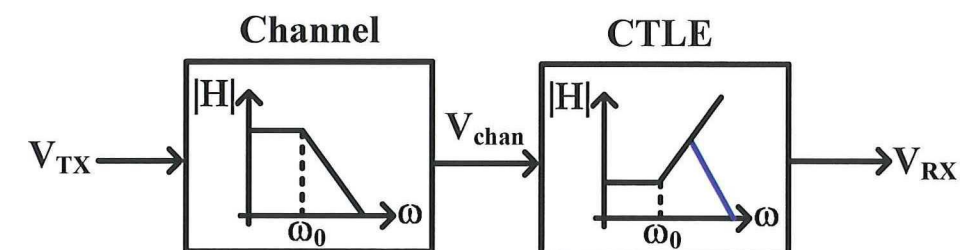
DFE Design: Closing the First Tap



- First tap timing often the most challenging
 - Other taps already have digital inputs
 - First tap must resolve/amplify small analog input signal
 - Larger " t_{dig} "

Continuous Time Linear Equalizer (CTLE)

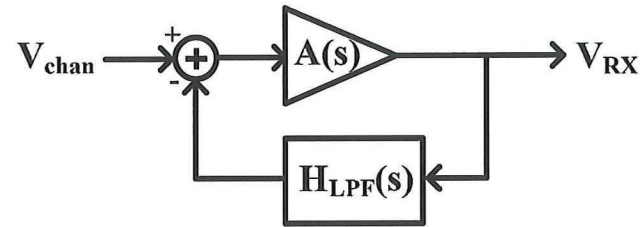
- Basic idea: build an analog circuit that approximates the inverse of the channel



- Can't (and don't need to) exactly invert – **finite gain @ high f**
 - CTLE bandwidth set by data-rate: typically $\sim 2/3 \cdot (1/T_{bit})$

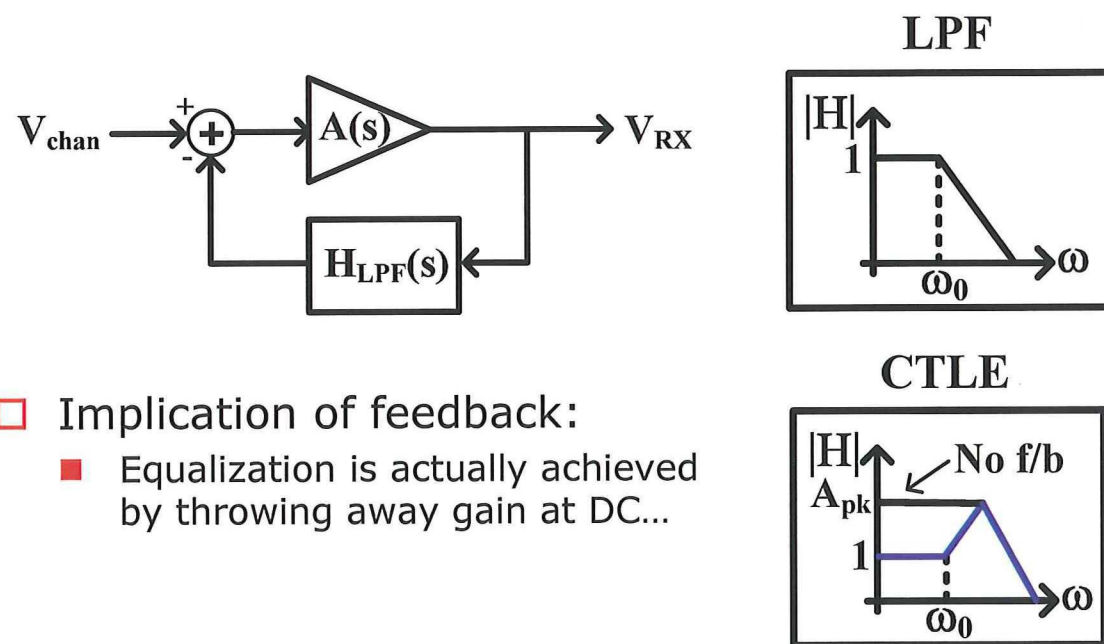
CTLE Implementation, Limitations (1)

- Often implemented using implicit or explicit negative feedback:



- Low-pass filter usually built out of R's and C's
 - Easy & efficient way to implement large time constants
 - Most effective with small # of poles & zeros
- May be hard to match complicated channels
 - Usually combined with a discrete-time equalizer

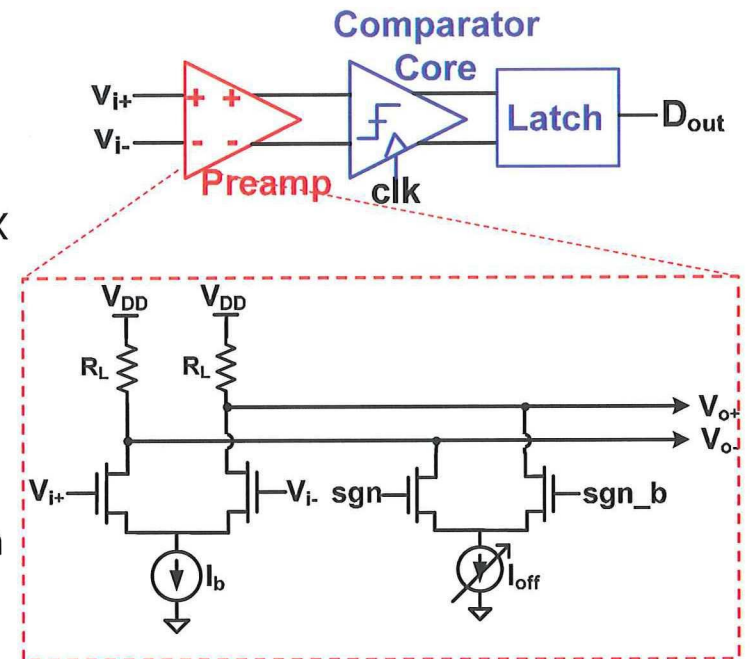
CTLE Implementation, Limitations (2)



- Implication of feedback:
 - Equalization is actually achieved by throwing away gain at DC...

Comparator Offset Cancellation

- Never fight offset with sizing
 - 4X penalty in capacitance and hence power for 2X reduction in offset
- Preamplifier is a convenient point to cancel offset
 - Keeps cancellation DAC parasitics off of the signal path



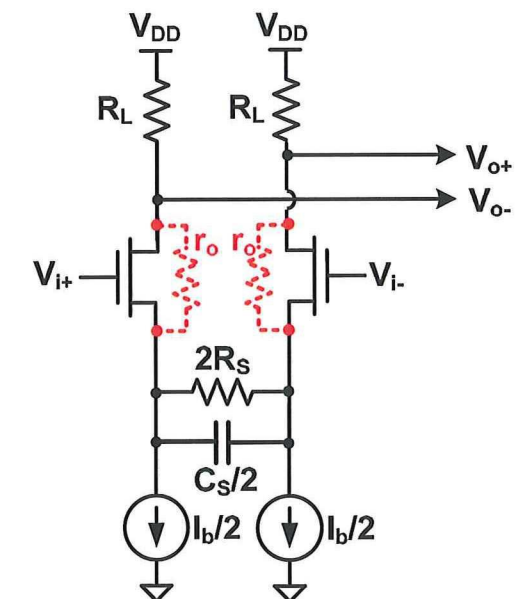
CTLE Design Issues (1)

Cause:

- Finite input device r_o

Effects:

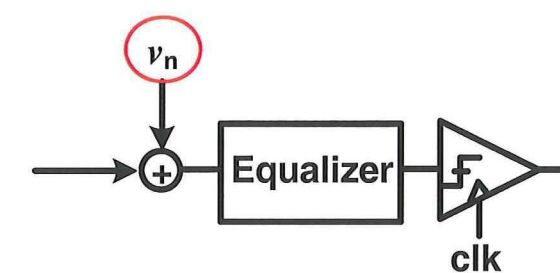
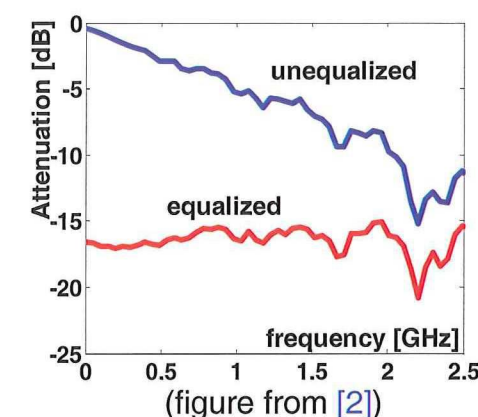
- Couples source and drain dynamics
- Makes precise setting of poles/zeros more difficult



Example DFE Power Numbers

- 20Gb/s 10-tap DFE in 65nm (same process/bias as preamp example)
 - $M_{DFE} = 2.5$, $V_{cursor} = 75\text{mV}$, $t_{dig} = 25\text{ps}$,
 $A_v = 1$, $C_L = 30\text{fF}$
- Current integration summer power: $\sim 337\mu\text{A}$
- Digital power (flip-flops, XORs, etc.): $\sim 2\text{mA}$ w/1V V_{DD}
 - Typically $\sim 10\text{fF}$ switching cap/tap

The Fundamental Issue: Noise



- Linear equalizers can eliminate ISI
 - But increase magnitude of noise (and high- f crosstalk) relative to signal
- Particularly bad on channels with notches
 - Equalizer needs large atten./gain

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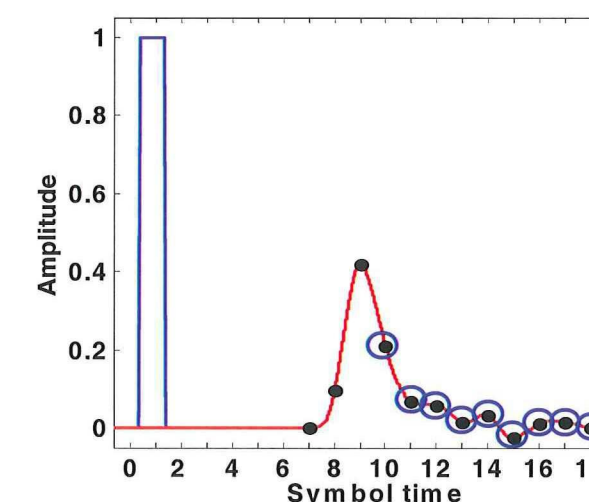
15

Tutorial Outline

- RX AFE Basics
- RX AFE Circuits & Power Analysis
- Practical Issues and Common Gotchas
- Key Lessons

Good News: There is Another Way

- Once you know which bit was transmitted
 - You also know exactly what ISI that bit will cause
- Why not directly cancel the ISI you know is coming?



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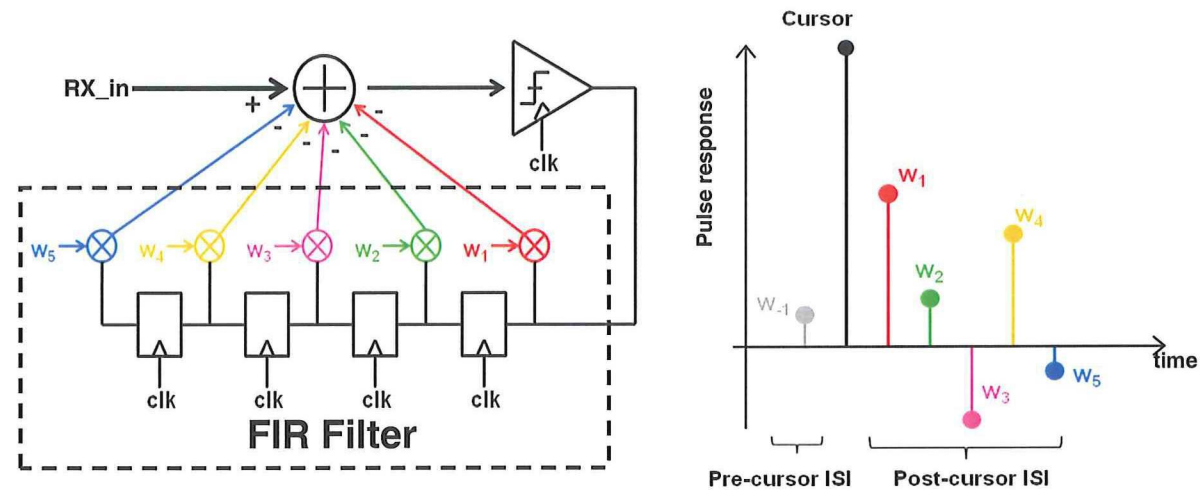
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16

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Decision Feedback Equalization



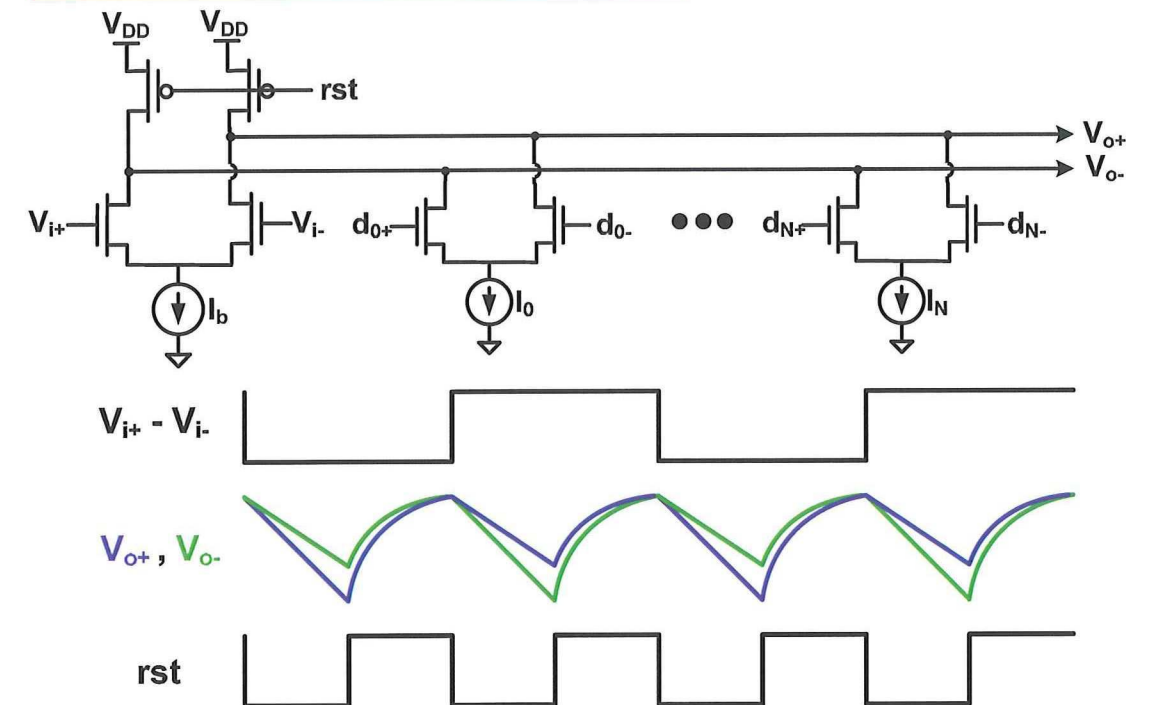
- Key advantage: no noise penalty
 - Feedback is based on "perfect" digital bits
 - ISI subtracted based on those bits

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17

Current Integrating Summer^[5]

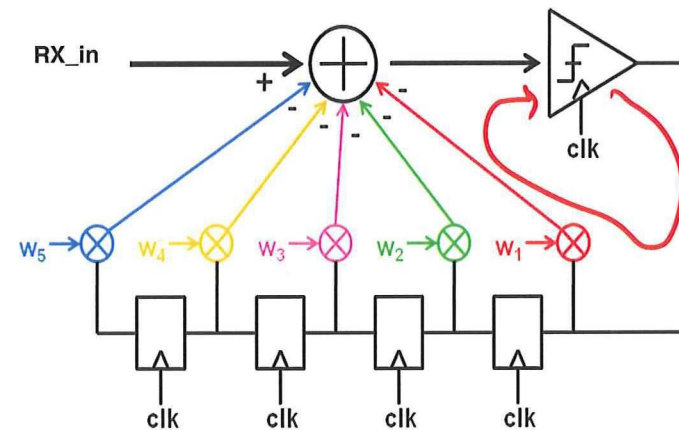


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35

Key Constraint: Timing



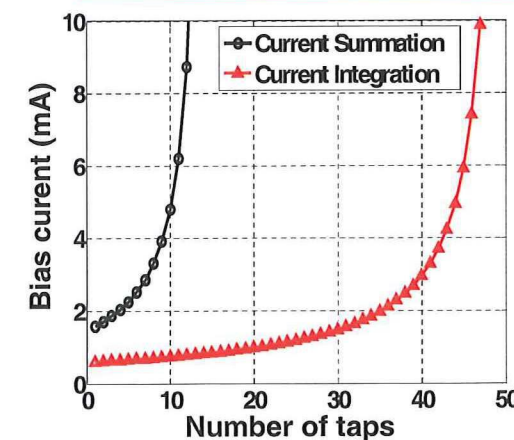
- Need to do all of the following in **at most 1 T_{bit}** :
 - Resolve the (small) input voltage
 - Scale the bit by the coefficient
 - Sum the new analog value

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18

Why Current Integration Helps

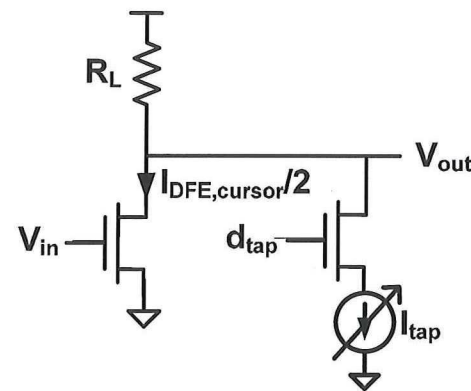


- Maximizes load impedance
 - ISI eliminated by resetting
- Power/number of taps improved by N_t :

$$I_{DFE, cursor} = \frac{A_v V^* C_L / (T_{bit} - t_{dig})}{1 - \left(1 + \frac{2M_{DFE} V_{cursor}}{V^*} \right) \frac{A_v / (T_{bit} - t_{dig})}{(\omega_T / \gamma)}}$$

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DFE Summer Power Consumption



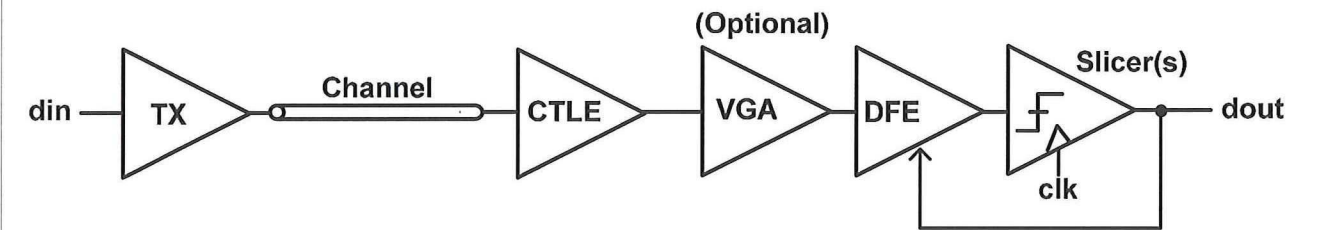
□ Definition:

$$M_{DFE} = \frac{V_{out,tap}}{V_{out,cursor}} = \frac{I_{tap}}{g_m V_{cursor}}$$

□ Assuming same biasing for tap & cursor devices (see [4]):

$$I_{DFE,cursor} = \frac{A_v V^* C_L N_\tau / (T_{bit} - t_{dig})}{1 - \left(1 + \frac{2M_{DFE} V_{cursor}}{V^*} \right) \frac{A_v N_\tau / (T_{bit} - t_{dig})}{(\omega_T / \gamma)}}$$

Complete Signal Chain



□ VGA sometimes included to ensure DFE processes its input linearly

- So that a linear FIR filter can be used to cancel the ISI
- More later

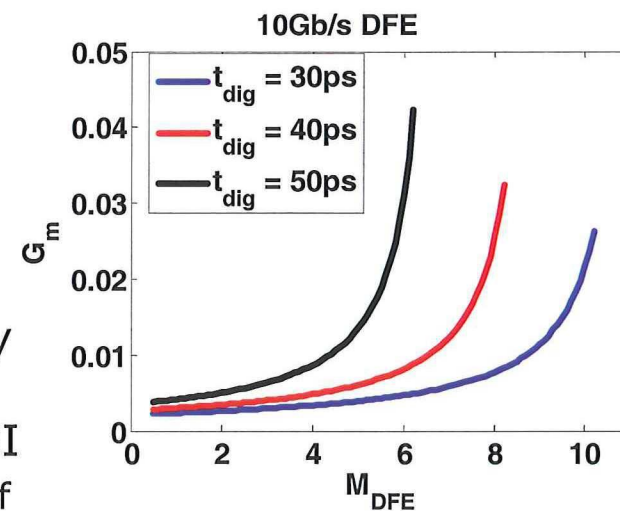
Implication

□ There is a maximum M_{DFE} you can achieve

□ Don't forget that every tap has to handle worst-case possible ISI

- Implies max. number of taps (in given tech.)

□ What if you need more taps/larger M_{DFE} ?



Tutorial Outline

□ RX AFE Basics

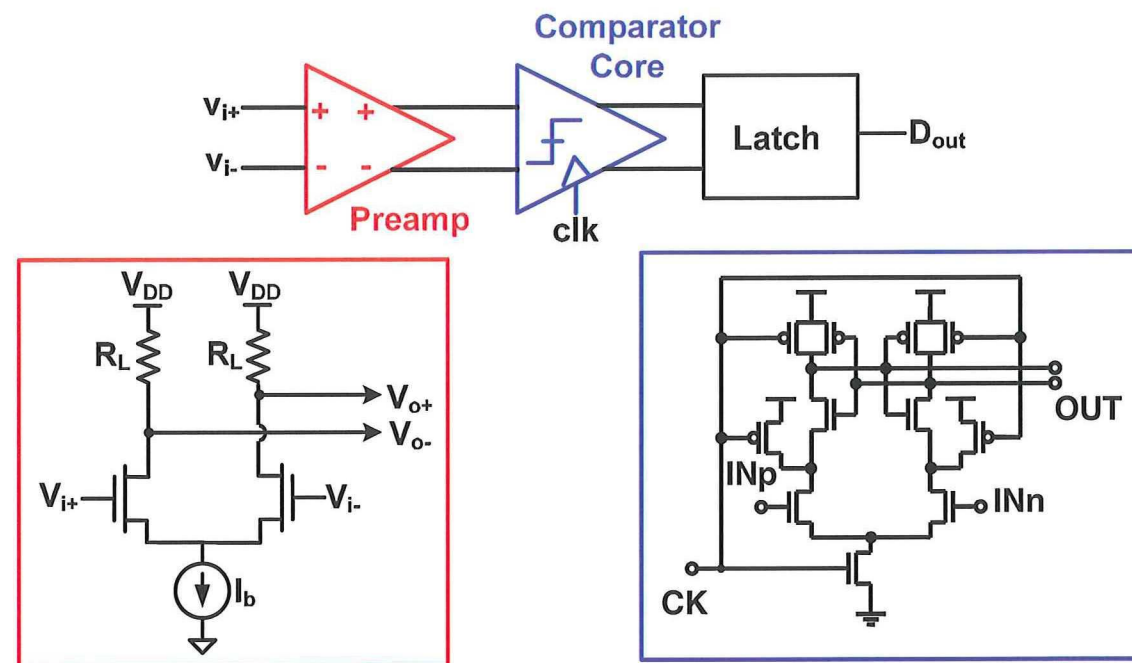
□ RX AFE Circuits & Power Analysis

- Comparator
- CTLE
- VGA
- DFE

□ Practical Issues and Common Gotchas

□ Key Lessons

Comparator Circuit Design



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21

DFE Summer Design

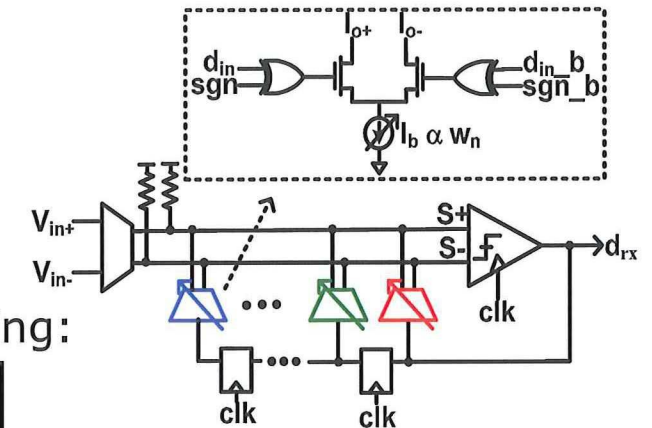
- Key constraint:

$$t_{dig_fb} + t_{ana_settle} = T_{bit}$$

- Need some N_τ of settling:

$$\tau_{sum} = (T_{bit} - t_{dig_fb})/N_\tau$$

- Typically want $\sim 4\tau$ for $\sim 98\%$ settling

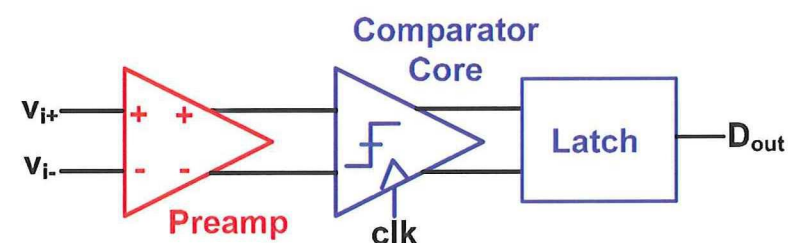


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31

Comparator Power Consumption



- Comparator core and latch typically just like digital gates:

$$P_{core,latch} = C_{sw,tot} V_{DD}^2 f_{clk}$$

- Let's look more carefully at preamp power

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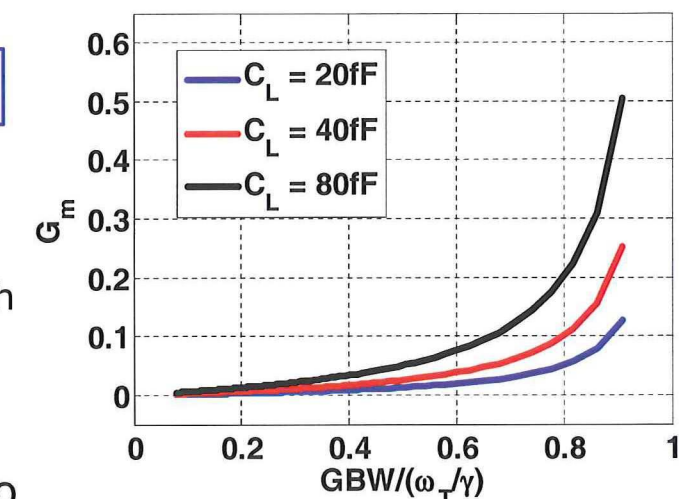
22

A Reminder: Self-Loading

$$I_{bias} = \frac{A_V \omega_{bw} V^* C_L}{1 - A_V \omega_{bw} / (\omega_T / \gamma)}$$

- Creates bandwidth limit
 - No matter how much power you spend

- DFE has increased "self-loading" due to taps...

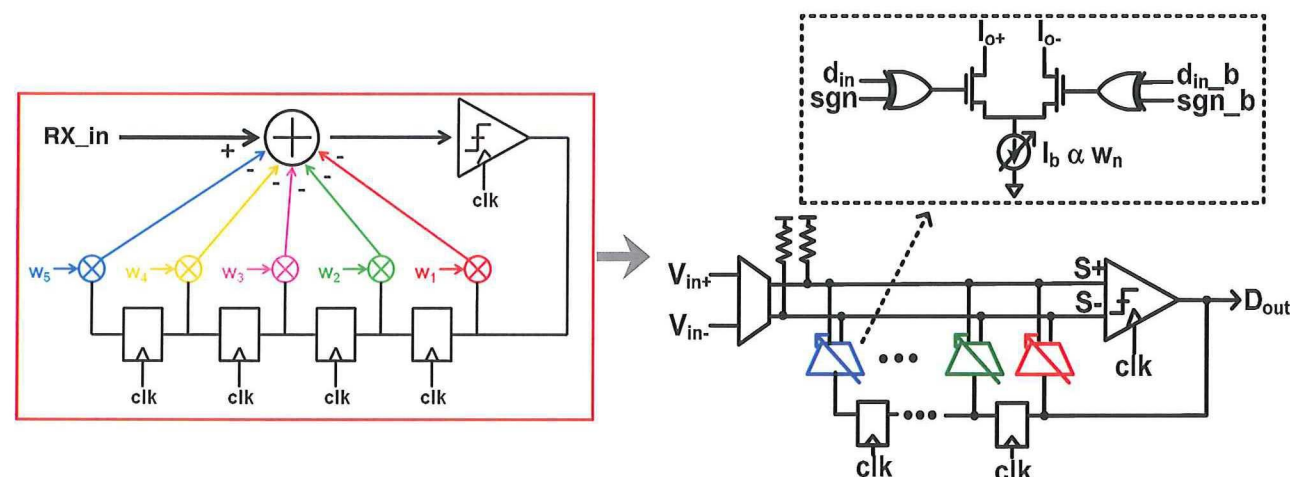


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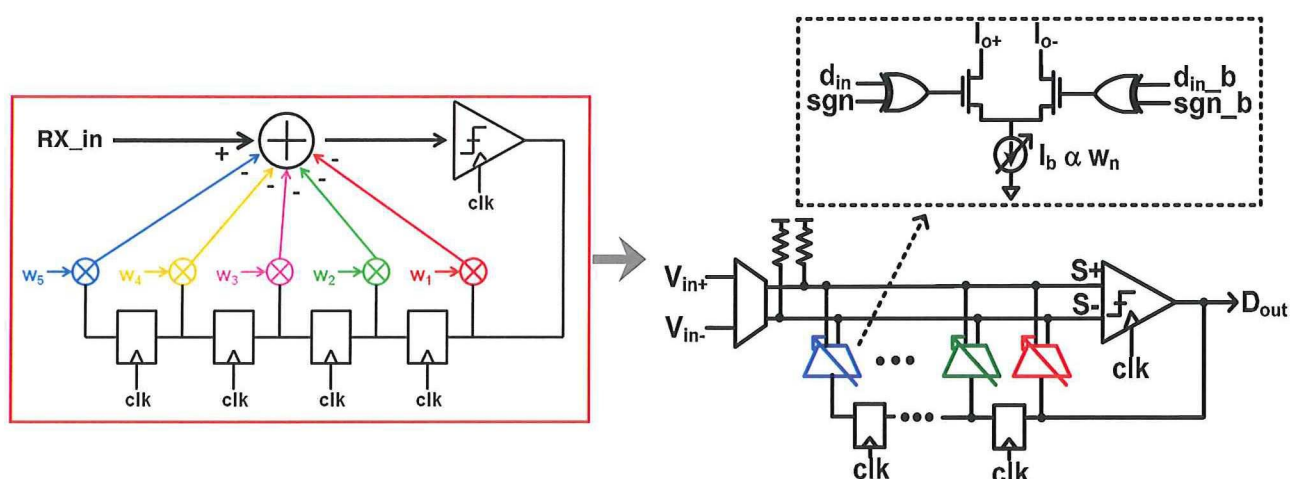
32

DFE Implementation (1)



- Mixed-signal FIR filter:
 - Digital delay
 - Analog scaling/summation

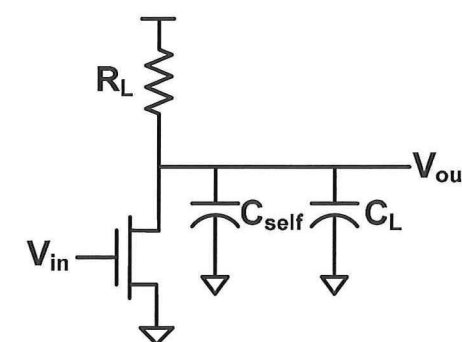
DFE Implementation (2)



- Design driven by feedback latency constraint
 - Summer output must settle within one bit time

Preamp Power Consumption (1)

- Single-ended equiv. circuit:
- Reminder and definition:



$$\frac{g_m}{C_{self} + C_L} = A_V \omega_{bw}$$

$$C_{self} = \gamma \frac{g_m}{\omega_T}$$

- So:

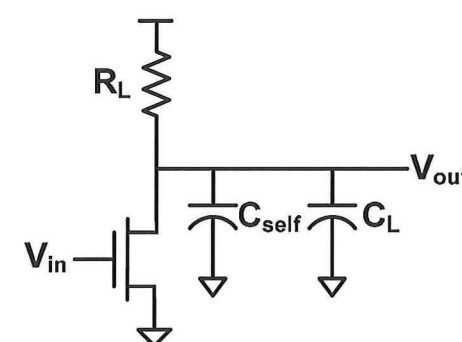
$$g_m = \frac{A_V \omega_{bw} C_L}{1 - A_V \omega_{bw} / (\omega_T / \gamma)}$$

Preamp Power Consumption (2)

- Single-ended equiv. circuit:

$$g_m = \frac{A_V \omega_{bw} C_L}{1 - A_V \omega_{bw} / (\omega_T / \gamma)}$$

$$V^* \equiv \frac{I_D}{g_m} \text{ (diff. pair)}$$

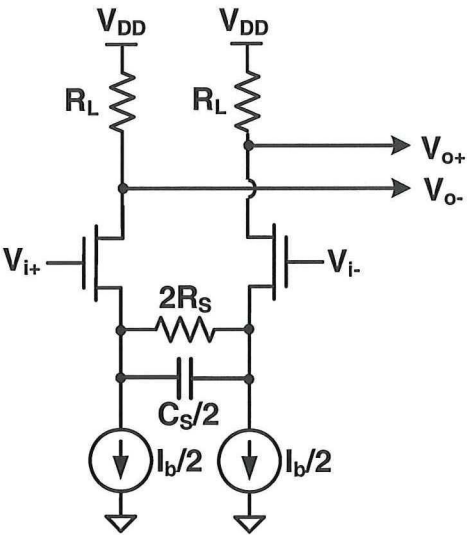


$$I_{bias} = \frac{A_V \omega_{bw} V^* C_L}{1 - A_V \omega_{bw} / (\omega_T / \gamma)}$$

- Typical 20Gb/s numbers in 65nm:
 - $A_V = 2$, $\omega_{bw} = 80\text{Grps}$, $V^* = 200\text{mV}$, $C_L = 10\text{fF}$, $\omega_T = 400\text{Grps}$
 - $I_{bias} = 533\mu\text{A}$

CTLE Circuit Design

- Most common design: source-degenerated
 - Excellent linearity – more later



- At system level, really only care about:
 - ω_{zero} ← set by channel pole(s)
 - ω_{bw} ← CTLE BW (data-rate)
 - A_{pk} ← peak gain
 - A_{DC} ← DC gain

CTLE Power Consumption

- Identical equation as preamp:

$$I_{bias} = \frac{A_{pk} \omega_{bw} V^* C_L}{1 - A_{pk} \omega_{bw} / (\omega_T / \gamma)}$$

- (Remember, CTLE is just an amplifier with gain reduced at low frequencies)

CTLE Design Equations (Simplified)

System	Circuit
$A_{pk} = g_m R_L$	$g_m = A_{pk} \omega_{bw} C_L$
$\omega_{bw} = \frac{1}{R_L C_L}$	$R_L = \frac{1}{\omega_{bw} C_L}$
$\frac{A_{pk}}{A_{DC}} = (1 + g_m R_s)$	$R_s = \left(\frac{A_{pk}}{A_{DC}} - 1 \right) \frac{1}{g_m}$
$\omega_{zero} = \frac{1}{R_s C_s}$	$C_s = \frac{g_m}{\omega_{zero} (A_{pk} - A_{DC})}$

VGA Implementation and Power

- Many possible implementations

- Source degeneration again attractive:
 - Better linearity when gain is low
 - (Ideally) no extra loading on signal path

- Equation for power once again identical to preamp
 - Set by max. achievable gain

